

BSCDS46: RESEARCH METHODOLOGY**BSc Data Science. IV Year VII Sem.**

Teaching Scheme: TH: - 4 Hours/Week	Credit TH: 04	Examination Scheme: In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 100 Marks Total : 100 Marks
Course Prerequisites: Basic understanding of analytics and mathematics		
Course Objective: The aim of this course is basic understanding of mathematical modeling, gap identification, study existing system, proposed new system for systematic research work.		
Learning Course Outcome: At the end of the course the students will be able to CO1: Learn to focus on a research problem using scientific methods. CO2: Learn methods to devise and design an experimentation set-up CO3: Learn basic instrumentation and data collection methods. CO4: Learn parameter estimation and related modelling methods		
Course Contents		
UNIT-I	Introduction Evolution Of Research	12 Hours
Evolution of Research Methodology: Meaning, nature, scope, and significance of research; Research paradigm; The purpose and outcomes of Research; Objectives of research, Motivation for research; Postulates underlying scientific investigations; Types of research; Research process and work flow. Engineering Research-Why? Research Questions, Engineering Ethics, conclusive proof-what constitutes, A research project-Why take on? Case Study- Code of Ethics, IEEE Code of Ethics, ACM Software Engineering Code of Ethics and Professional Practice, Code of Ethics especially covering Engineering discipline, various aspects environment, sustainable outcomes, employer, public, and Nation, Engineering Disasters.		
UNIT-II	Archival Literature	12 Hours
Why should engineers be ethical? Types of publications- Journal papers, conference papers, books, standards, patents, theses, trade magazine, newspaper article, infomercials, advertisement, Wikipedia & websites, Measures of research impact, Literature review publication cost. Case Study- Engineering dictionary, Shodhganga, The Library of Congress, Research gate, Google Scholar, Bibliometrics, Citations, Impact Factor, h-index, I-index, plagiarism, copyright infringement. Developing Research Plan: Research Proposals, Finding a suitable research questions, The elements of research proposals-title, details, budget, Design. for outcomes-1D data, 2D data, 3D data, N-D data, The research tools- Experimental measurements, numerical modeling, theoretical derivations & Calculations, curve matching. Case Study- Various Research grants and funding resources		
UNIT-III	Statistical Analysis:	12 Hours

Introduction, Sources of error and uncertainty, One-Dimensional Statistics: combining errors and uncertainties, t-test, ANOVA statistics, example, Two-Dimensional Statistics: example, Multi-Dimensional Statistics: partial correlation coefficients, example, Null hypothesis testing. Case Study- GNU PSPP Tool, SOFA, NOST-Dataplot.		
UNIT-IV	Optimization Techniques:	12 Hours
Introduction, Two-parameter optimization methods: sequential uniform sampling, Monte Carlo optimization, Simplex Optimization method, Gradient Optimization method, multi-parameter optimization methods, The cost function. Case Study- Google Optimization Tools, OpenMDAO		
UNIT-V	Survey Research Methods:	12 Hours
Why undertake a survey, Ergonomics and human factors, Ethics approval, General survey guidelines, Survey statements, Survey delivery, Respondent selection, Survey timelines, Statistical analysis, Reporting. Case Study- Qualitative Analysis Tools- AQUAD, CAT, Research presentation: Introduction, Standard terms, Standard research methods and experimental techniques, Paper title and keywords, Writing an abstract, Paper presentation and review, Conference presentations, Poster presentations, IPR, Copyright, Patents. Reporting Research: Thesis, Structure and Style for writing thesis, Dissemination of research findings; Reporting and interpretation of results; cautions in interpretations, Type of reports, Typical report outlines. The path forward: Publication trends,		
TEXTBOOKS: 1. David V Thiel, —Research Methods- for EngineersI, Cambridge University Press, ISBN:978-1-107-61019-4 2. Kothari C.R., —Research Methodology. New Age International, 2004, 2 nd Ed; ISBN:13: 978-81-224-1522-3.		
REFERENCES: 1. Caroline Whitbeck, —Ethics in Engineering Practice and ResearchI, 2nd Ed, Cambridge University Press; ISBN :978-1-107-66847-8 2. Gordana DODIG-CRNKOVIC, —Scientific Methods in Computer ScienceI, Department of Computer Science Malardalen University, Vasteas, Sweden; ISBN: 91-26-97860-1		